Homework 10 Tuesday

Sunday, 17 March 2024 5:00 pm

Homework 10

Exercise set 5.4

Q3, Q4, Q6, Q13, Q 15, Q 18, Q 25, Q 26, Q 33

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3. Which of the following sets of vectors are bases for R^3?

(a) (1, 0, 0), (2, 2, 0), (3, 3, 3)

(b) (3, 1, −4), (2, 5, 6), (1, 4, 8)

(c) (2, 3, 1), (4, 1, 1), (0, −7, 1)

(d) (1, 6, 4), (2, 4, −1), (−1, 2, 5)

(a)

We need to check linear independence and spanning of the given vectors i.e. we need to solve two systems

$$c_1(1,0,0)+c_2(2,2,0)+c_3(3,3,3)=(0,0,0) \quad c_1(1,0,0)+c_2(2,2,0)+c_3(3,3,3)=(b_1,b_2,b_3)$$

$$\begin{aligned} c_1+2c_2+3c_3&=0\\ 2c_2+3c_3&=0\\ 3c_3&=0 \end{aligned}$$

$$\begin{aligned} c_1+2c_2+3c_3&=b_1\\ 2c_2+3c_3&=b_2\\ 3c_3&=b_3 \end{aligned}$$

If we prove that $\det A \neq 0$, where

$$A=\begin{bmatrix}1&2&3\\0&2&3\\0&0&3\end{bmatrix}$$

(note that columns of matrix A are coordinates of given vectors), we will prove that homogeneous system has trivial solution and nonhomogeneous system has solution i.e. given vectors will form basis for $\mathbb{R}^3$. Let's compute determinant

$$\begin{vmatrix}1&2&3\\0&2&3\\0&0&3\end{vmatrix}=1\cdot2\cdot3=6\neq0$$

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4. Which of the following sets of vectors are bases for P_2?

(a) $1-3x+2x^2,\ 1+x+4x^2,\ 1-7x$

(b) $4+6x+x^2,\ -1+4x+2x^2,\ 5+2x-x^2$

(c) $1+x+x^2,\ x+x^2,\ x^2$

(d) $-4+x+3x^2,\ 6+5x+2x^2,\ 8+4x+x^2$

(a)

We need to check linear independence and spanning of the given vectors i.e. we need to solve two systems

$$c_1(1-3x+2x^2)+c_2(1+x+4x^2)+c_3(1-7x)=0$$

$$c_1(1-3x+2x^2)+c_2(1+x+4x^2)+c_3(1-7x)=b_1+b_2x+b_3x^2$$

$$\begin{aligned} c_1+c_2+c_3&=0\\ -3c_1+c_2-7c_3&=0\\ 2c_1+4c_2&=0 \end{aligned}$$

$$\begin{aligned} c_1+c_2+c_3&=b_1\\ -3c_1+c_2-7c_3&=b_2\\ 2c_1+4c_2&=b_3 \end{aligned}$$

If we prove that $\det A\neq 0$, where

$$A=\begin{bmatrix} 1 & 1 & 1\\ -3 & 1 & -7\\ 2 & 4 & 0 \end{bmatrix}$$

(note that columns of matrix A are coefficients of given vectors), we will prove that homogeneous system has trivial solution and nonhomogeneous system has solution i.e. given vectors will form basis for P_2. Let's compute determinant

$$\begin{aligned} \det A&=\begin{vmatrix} 1 & 1 & 1\\ -3 & 1 & -7\\ 2 & 4 & 0 \end{vmatrix}\\ &=2\begin{vmatrix} 1 & 1\\ 1 & -7 \end{vmatrix}-4\begin{vmatrix} 1 & 1\\ -3 & -7 \end{vmatrix}\\ &=2(-7-1)-4(-7+3)\\ &=0 \end{aligned}$$

Since $\det A=0$, given vectors don't form basis for P_2.

(b)

We will use procedure from part (a)

$$\begin{aligned} \det A&=\begin{vmatrix} 4 & -1 & 5\\ 6 & 4 & 2\\ 1 & 2 & -1 \end{vmatrix}\\ &=\begin{vmatrix} -1 & 5\\ 4 & 2 \end{vmatrix}-2\begin{vmatrix} 4 & 5\\ 6 & 2 \end{vmatrix}-\begin{vmatrix} 4 & -1\\ 6 & 4 \end{vmatrix}\\ &=-2-20-2(8-30)-(16+6)\\ &=0 \end{aligned}$$

Since $\det A=0$, given vectors don't form basis for P_2.

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6. Let V be the space spanned by $\mathbf{v}_1 = \cos^2 x$, $\mathbf{v}_2 = \sin^2 x$, $\mathbf{v}_3 = \cos 2x$.

(a) Show that $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is not a basis for V.

(b) Find a basis for V.

Step 1

Let V be the space spanned by $\mathbf{v}_1 = \cos^2 x$, $\mathbf{v}_2 = \sin^2 x$ and $\mathbf{v}_3 = \cos 2x$.

(a) Observe that $v_1 - v_2 - v_3 = \cos^2 x - \sin^2 x - \cos 2x = \cos^2 x - \sin^2 x - \left(\cos^2 x - \sin^2 x\right) = \cos^2 x - \sin^2 x - \cos^2 x + \sin^2 x = 0$ for all $x \in (-\infty, \infty)$ which gives us that the set $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a linearly dependent set on V and therefore S is not a basis for V.

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determine the dimension of and a basis for the solution space of the system.

13.
$$\begin{aligned} 3x_1 + x_2 + x_3 + x_4 &= 0 \\ 5x_1 - x_2 + x_3 - x_4 &= 0 \end{aligned}$$

Step 1

The given system can be expressed in the form $Ax = 0$ where,

$$A = \begin{bmatrix} 3 & 1 & 1 & 1 \\ 5 & -1 & 1 & -1 \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

The matrix A can be reduced as

$$\begin{aligned} A &= \begin{bmatrix} 3 & 1 & 1 & 1 \\ 5 & -1 & 1 & -1 \end{bmatrix} \\ C_1 \leftrightarrow C_2 &: \begin{bmatrix} 1 & 3 & 1 & 1 \\ -1 & 5 & 1 & -1 \end{bmatrix} \\ R_2 \to R_2 + R_1 &: \begin{bmatrix} 1 & 3 & 1 & 1 \\ 0 & 8 & 2 & 0 \end{bmatrix} \end{aligned}$$

Thus, the given system becomes

$$\begin{bmatrix} 1 & 3 & 1 & 1 \\ 0 & 8 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 + 3x_2 + x_3 + x_4 = 0$$
$$8x_2 + 2x_3 = 0$$
$$\implies x_3 = -4x_2$$
$$x_1 + 3x_2 - 4x_2 + x_4 = 0$$
$$x_1 - x_2 + x_4 = 0 \implies x_1 \quad = x_2 - x_4$$

$$x = \begin{bmatrix} x_2 - x_4 \\ x_2 \\ -4x_2 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \\ -4 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

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Step 2

Since the vectors $(1,1,-4,0), (-1,0,0,1)$ are linearly independent in $\mathbb{R}^4$ as they are not multiple of each other. So dimension of solution space is 2 and its basis is given by $\{(1,1,-4,0),(-1,0,0,1)\}$.

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15.
$$\begin{aligned} x_1 - 3x_2 \quad + x_3 &= 0 \\ 2x_1 - 6x_2 + 2x_3 &= 0 \\ 3x_1 - 9x_2 + 3x_3 &= 0 \end{aligned}$$

Step 1

Note that the given system of homogeneous equation can be written as $Ax = \mathbf{0}$, where

$$A = \begin{pmatrix} 1 & -3 & 1 \\ 2 & -6 & 2 \\ 3 & -9 & 3 \end{pmatrix} \text{ and } x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

Let us row reduce the matrix A, before proceeding further. We have

$$\begin{aligned} A &= \begin{pmatrix} 1 & -3 & 1 \\ 2 & -6 & 2 \\ 3 & -9 & 3 \end{pmatrix} \\ &\sim \begin{pmatrix} 1 & -3 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{(applying the operations } R_2 - 2R_1, R_3 - 3R_1). \end{aligned}$$

Hence we have that the system of equation reduces to

$$\begin{aligned} &\begin{pmatrix} 1 & -3 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\ &\implies x_1 - 3x_2 + x_3 = 0 \\ &\implies x_3 = -x_1 + 3x_2 \end{aligned}$$

Hence the solution set is

$$\begin{aligned} &\{(x_1, x_2, x_3) : x_3 = -x_1 + 3x_2, x_i \in \mathbb{R}\} \\ &= \{(x_1, x_2, -x_1 + 3x_2) : x_1, x_2 \in \mathbb{R}\} \\ &= \{x_1(1, 0, -1) + x_2(0, 1, 3) : x_1, x_2 \in \mathbb{R}\}, \end{aligned}$$

which is two dimensional and a basis for the same is given by $\{(1, 0, -1), (0, 1, 3)\}$, as they are linearly independent.

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18. Determine bases for the following subspaces of R^3.

(a) the plane $3x - 2y + 5z = 0$

(b) the plane $x - y = 0$

(c) the line $x = 2t$, $y = -t$, $z = 4t$

(d) all vectors of the form (a, b, c), where $b = a + c$

Step 1

Let us denote the given subspace by W. Then we have

$$\begin{aligned} W &= \{(x,y,z) \in \mathbb{R}^3 : 3x - 2y + 5z = 0\} \\ &= \{(x,y,z) \in \mathbb{R}^3 : 2y = 3x + 5z\} \\ &= \left\{(x,y,z) \in \mathbb{R}^3 : y = \frac{3}{2}x + \frac{5}{2}z\right\} \\ &= \left\{\left(x, \frac{3}{2}x + \frac{5}{2}z, z\right) : x, z \in \mathbb{R}\right\} \\ &= \left\{x\left(1, \frac{3}{2}, 0\right) + z\left(0, \frac{5}{2}, 1\right) : x, z \in \mathbb{R}\right\}. \end{aligned}$$

$$\begin{aligned} & k_1\left(1, \frac{3}{2}, 0\right) + k_2\left(0, \frac{5}{2}, 1\right) = (0,0,0) \\ \Longrightarrow & \left(k_1, \frac{3}{2}k_1 + \frac{5}{2}k_2, k_2\right) = (0,0,0) \\ \Longrightarrow & k_1 = k_2 = 0. \end{aligned}$$

The set $\left\{\left(1, \frac{3}{2}, 0\right), \left(0, \frac{5}{2}, 1\right)\right\}$ is a basis for the given subspace.

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Step 1

Let us denote the given subspace by W. Then we have

$$\begin{aligned} W &= \{(x,y,z) \in \mathbb{R}^3 : x - y = 0\} \\ &= \{(x,y,z) \in \mathbb{R}^3 : x = y\} \\ &= \{(x,x,z) : x, z \in \mathbb{R}\} \\ &= \{x(1,1,0) + z(0,0,1) : x, z \in \mathbb{R}\}. \end{aligned}$$

$$\begin{aligned} & k_1(1,1,0) + k_2(0,0,1) = (0,0,0) \\ \Longrightarrow & (k_1, k_1, k_2) = (0,0,0) \\ \Longrightarrow & k_1 = k_2 = 0. \end{aligned}$$

Result

The set $\{(1,1,0), (0,0,1)\}$ is a basis for the given subspace.

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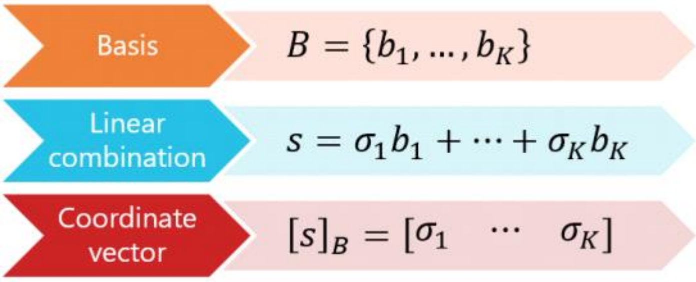

In linear algebra, a **coordinate vector** is a representation of a vector as an ordered list of numbers (a tuple) that describes the vector in terms of a particular ordered basis.[1] An easy example may be a position such as (5, 2, 1) in a 3-dimensional Cartesian coordinate system with the basis as the axes of this system. Coordinates are always specified relative to an ordered basis. Bases and their associated coordinate representations let one realize vector spaces and linear transformations concretely as column vectors, row vectors, and matrices; hence, they are useful in calculations.

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Example 1 [edit]

Let P_3 be the space of all the algebraic polynomials of degree at r
the following polynomials:

$$B_P = \{1, x, x^2, x^3\}$$

matching

$$1 := \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}; \quad x := \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}; \quad x^2 := \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}; \quad x^3 := \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

then the coordinate vector corresponding to the polynomial

$$p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$$

is

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}.$$

25. Let S be a basis for an n-dimensional vector space V. Show that if $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r$ form a linearly independent set of vectors in V, then the coordinate vectors $(\mathbf{v}_1)_S, (\mathbf{v}_2)_S, \ldots, (\mathbf{v}_r)_S$ form a linearly independent set in R^n, and conversely.

25. First notice that if $\mathbf{v}$ and $\mathbf{w}$ are vectors in V and a and b are scalars, then $(a\mathbf{v} + b\mathbf{w})_S = a(\mathbf{v})_S + b(\mathbf{w})_S$. This follows from the definition of coordinate vectors. Clearly, this result applies to any finite sum of vectors. Also notice that if $(\mathbf{v})_S = (\mathbf{0})_S$, then $\mathbf{v} = \mathbf{0}$. Why?

Now suppose that $k_1\mathbf{v}_1 + \cdots + k_r\mathbf{v}_r = \mathbf{0}$. Then

$$\begin{aligned}(k_1\mathbf{v}_1 + \cdots + k_r\mathbf{v}_r)_S &= k_1(\mathbf{v}_1)_S + \cdots + k_r(\mathbf{v}_r)_S \\ &= (\mathbf{0})_S\end{aligned}$$

Conversely, if $k_1(\mathbf{v}_1)_S + \cdots + k_r(\mathbf{v}_r)_S = (\mathbf{0})_S$, then

$$(k_1\mathbf{v}_1 + \cdots + k_r\mathbf{v}_r)_S = (\mathbf{0})_S, \quad \text{or} \quad k_1\mathbf{v}_1 + \cdots + k_r\mathbf{v}_r = \mathbf{0}$$

Thus the vectors $\mathbf{v}_1, \ldots, \mathbf{v}_r$ are linearly independent in V if and only if the coordinate vectors $(\mathbf{v}_1)_S, \ldots, (\mathbf{v}_r)_S$ are linearly independent in R^n.

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Step 1

Let $\{v_1, v_2, \cdots, v_r\}$ be a linearly independent set of vectors of V. If possible assume $\{(v_1)_S, (v_2)_S, \cdots, (v_r)_S\}$ to be linearly dependent. Then there exists $k_1, k_2, \cdots, k_r \in \mathbb{R}$, not all zero such that

$$\begin{aligned}
& k_1(v_1)_S + k_2(v_2)_S + \cdots + k_r(v_r)_S = 0 \\
\implies & (k_1v_1 + k_2v_2 + \cdots + k_rv_r)_S = 0_S \qquad (\text{as } k(v)_S = (kv)_S, \\
& \qquad k(v)_S + l(w)_S = (kv+lw)_S, k, l \in \mathbb{R}) \\
\implies & k_1v_1 + k_2v_2 + \cdots + k_rv_r = 0 \\
\implies & k_1 = k_2 = \cdots = k_r = 0 \qquad (\text{as } \{v_1, v_2, \cdots, v_r\} \text{ is linearly independent}).
\end{aligned}$$

Hence $\{(v_1)_S, (v_2)_S, \cdots, (v_r)_S\}$ is linearly independent.

Conversely suppose $\{(v_1)_S, (v_2)_S, \cdots, (v_r)_S\}$ is linearly independent and $\{v_1, v_2, \cdots, v_r\}$ be linearly dependent. Then there exist $k_1, k_2, \cdots, k_r \in \mathbb{R}$, not all zero such that

$$\begin{aligned}
& k_1v_1 + k_2v_2 + \cdots + k_rv_r = 0 \\
\implies & (k_1v_1 + k_2v_2 + \cdots + k_rv_r)_S = 0_S \\
\implies & (k_1v_1)_S + (k_2v_2)_S + \cdots + (k_rv_r)_S = 0_S \\
\implies & k_1(v_1)_S + k_2(v_2)_S + \cdots + k_r(v_r)_S = 0_S \\
\implies & k_1 = k_2 = \cdots = k_r = 0 \qquad (\text{as } \{(v_1)_S, (v_2)_S, \cdots, (v_r)_S\} \text{ is linearly independent}).
\end{aligned}$$

26. Using the notation from Exercise 25, show that if $v_1, v_2, \ldots, v_r$ span V, then the coordinate vectors $(v_1)_S, (v_2)_S, \ldots, (v_r)_S$ span R^n, and conversely.

Let $v_1, v_2, \cdots, v_r$ span V and $w \in \mathbb{R}^n$. Since S is a basis of V, we have that there exists $v \in V$, such that $(v)_S = w$. Since $v_1, v_2, \cdots, v_r$ span V, we have $k_1, k_2, \cdots, k_r \in \mathbb{R}$, such that $k_1v_1 + k_2v_2 + \cdots + k_rv_r = v$. Hence we have

$$\begin{aligned}
& k_1v_1 + k_2v_2 + \cdots + k_rv_r = v \\
\implies & (k_1v_1 + k_2v_2 + \cdots + k_rv_r)_S = (v)_S \\
\implies & k_1(v_1)_S + k_2(v_2)_S + \cdots + k_r(v_r)_S = w.
\end{aligned}$$

Hence $\{(v_1)_S, (v_2)_S, \cdots, (v_r)_S\}$ spans $\mathbb{R}^n$.

Conversely assume $(v_1)_S, (v_2)_S, \cdots, (v_r)_S$ spans $\mathbb{R}^n$ and $v \in V$. Then $(v)_S = w \in \mathbb{R}^n$. Hence there exis $k_1, k_2, \cdots, k_r \in \mathbb{R}$, such that $k_1(v_1)_S + k_2(v_2)_S + \cdots + k_r(v_r)_S = w$. Hence we have

$$\begin{aligned}
& k_1(v_1)_S + k_2(v_2)_S + \cdots + k_r(v_r)_S = (v)_S \\
\implies & (k_1v_1 + k_2v_2 + \cdots + k_rv_r)_S = (v)_S \\
\implies & k_1v_1 + k_2v_2 + \cdots + k_rv_r = v.
\end{aligned}$$

Hence $v_1, v_2, \cdots, v_r$ span V.

33.

(a) For a 3×3 matrix A, explain in words why the set $I_3, A, A^2, \ldots, A^9$ must be linearly dependent if the ten matrices are distinct.

(b) State a corresponding result for an $n \times n$ matrix A.

Step 1

Let us first prove that M_{33} has dimension 9. Note that for any

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \in M_{33}$$

, we have

$$\begin{aligned}
&\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \\
=&a\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + b\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + c\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
&+ d\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + e\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} + f\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \\
&+ g\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} + h\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} + i\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.
\end{aligned}$$

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Step 1

Just as done in $33(a)$, if we take for $1 \le i, j \le n$, the element E_{ij}, which has 1 at (i,j)-th position and 0 elsewhere, then the set $\mathcal{B} = \{E_{ij} : 1 \le i, j \le n\}$ spans M_{nn} and is linearly independent. Hence if the collection $\{I_n, A, A^2, \cdots, A^{n^2}\}$ has $n^2 + 1$ different elements, then it will be linearly dependent.

Result

2 o

We prove that if the collection $\{I_n, A, A^2, \cdots, A^{n^2}\}$ has $n^2 + 1$ different elements, then it will be linearly dependent

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